

PROBABILISTIC ROBOTICS LAB – KALMAN FILTER, EXTENDED KALMAN FILTER, PARTICLE FILTER PERFORMANCE EVALUATION UNDER VARYING PROCESS AND MEASUREMENT NOISE

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Abstract: Accurate state estimation is a fundamental prerequisite for autonomous mobile robot navigation. In this paper, the performance boundaries and noise resilience of four classic probabilistic state estimation methodologies are evaluated: a linear Kalman Filter (KF), an Extended Kalman Filter (EKF) with and without landmark detection, and a Particle Filter (PF). The frameworks were implemented as independent ROS2 nodes and evaluated within a unified Gazebo simulation environment using a TurtleBot3 platform. A rigorous testing protocol of 16 experimental runs was executed, systematically scaling the process and measurement noise Q and R across four orders of magnitude while the robot executed a path featuring five zero-radius turns on the spot. The results demonstrate that the linear KF suffers from permanent coordinate drift due to its structural inability to resolve angular state coupling. The EKF effectively handles non-linearities but exhibits localized instability and final endpoint degradation under extreme process noise configurations due to aggressive landmark-driven correction updates. The Particle Filter demonstrates the highest overall resilience against extreme sensor degradation by propagating its swarm via direct control commands instead of noisy wheel-encoder data, and it actively prevents severe tracking failures through a geometric plausibility check on coordinate jumps. To ensure full scientific verifiability and reproducibility, the source code and the trajectory plots are accessible in a Git repository.¹

1. INTRODUCTION

In mobile robotics, accurate state estimation remains a fundamental prerequisite for tasks such as path planning, navigation, and safe human-robot interaction [1-3]. While wheel odometry provides high-frequency motion data, it is inherently prone to cumulative drift caused by wheel slippage and manufacturing tolerances, demanding the integration of probabilistic state estimation techniques. This paper evaluates the performance of four state estimation architectures – a linear Kalman Filter (KF), an Extended Kalman Filter (EKF), an EKF with integrated landmark-detection, and a Particle Filter (PF)—implemented as native ROS2 nodes and validated using a custom-designed TurtleBot simulation equipped with distinct cylindrical landmarks in Gazebo. The primary contribution of this work lies in the empirical evaluation of these filters under the scaling of process noise Q and measurement noise R . This variation is critical to expose the limits of each estimator,

mapping the boundary conditions where the filter either over-relies on inaccurate kinematic models or succumbs to sensor noise, thereby establishing a benchmark for optimal filter tuning in mobile robotic platforms.

2. METHODS

To evaluate the state estimation architectures, a unified experimental framework was established in a Gazebo simulation environment using the TurtleBot3 Waffle. All filters operate within the same coordinate frames, utilize identical sensor data pipelines, and are subjected to the same test trajectories. This common system setup ensures that differences in estimated trajectories depend solely on the algorithmic properties and noise characteristics of the respective filters rather than variations in environmental factors.

2.1. Noisy Odometry

The baseline for kinematics-based state estimation is derived from the robot's wheel encoders. In ideal conditions, wheel odometry integrates differential wheel velocities to track position over time. However, real-world robotic systems encounter physical disturbances such as wheel slippage, surface irregularities, and mechanical misalignments. To mimic these real-world conditions within the simulation, the information published to the `/odom` topic is considered as the ground truth and a dedicated `joint_noise_node` injects zero-mean Gaussian noise directly into the joint states of the robot. The resulting corrupted data propagates through a `noisy_odometry_node`, which computes a degraded odometry estimate used by the implemented filters.

2.2. Kalman Filter

The first estimation architecture is a linear Kalman Filter. Unlike the other filters implemented in this work, this node does not estimate the robot's full pose directly on a 2D plane, as the trigonometric components of the unicycle kinematics introduce nonlinearities that a standard KF cannot handle [4]. Instead, the state vector is decoupled and defined strictly by the linear velocities along the axes and their corresponding Cartesian positions. The node subscribes to the degraded data from the noisy odometry network. It utilizes a linear, constant-velocity motion model to execute its prediction step, estimating where

¹ https://github.com/cedericoStudent/proLAB_CedericoV1

the robot should be based on its previous velocity state. During the update phase, the filter relies on global linear velocities extracted from the corrupted odometry topic. The measurement matrix is configured to isolate these velocity components, meaning that absolute position coordinates are never measured directly. Instead, the filter continuously corrects its linear velocity states and propagates these refinements to update the Cartesian positions. By decoupling the orientation angle from the state vector and omitting raw inertial measurements, the system dynamics remain linear.

2.3. Extended Kalman Filter

The second estimation architecture expands the state vector into a multi-variable representation containing the Cartesian positions and the robot orientation. Unlike the decoupled linear filter, this node directly addresses the nonlinearities of the TurtleBot's kinematics. The state vector is advanced during the prediction phase by integrating the local linear and angular velocities from the noisy odometry channel through trigonometric projection equations. To propagate the state uncertainty correctly, a Jacobian matrix is evaluated at each time step to linearize the nonlinear motion equations around the current state estimate. Additionally, the process noise matrix is defined in the local robot frame and rotated into the global coordinate frame at every iteration using a time-varying rotation matrix. For the measurement update, the filter fuses raw inertial data to bound the angular drift. The node subscribes directly to the IMU sensor topic and extracts the independent angular velocity around the vertical axis. Linear accelerations from the IMU are intentionally omitted from the measurement update, as minor surface irregularities project gravitational components onto the horizontal axes, and double-integrating noisy acceleration signals introduces a quadratic error drift that rapidly destabilizes position estimates [2]. Instead, the measurement matrix isolates the orientation state, allowing the filter to calculate an orientation residual based on the difference between the expected heading and the integrated IMU yaw rate. This angular residual is subsequently normalized to prevent wrapping issues before updating the state vector and its associated covariance matrix through the computed Kalman gain. While the linear velocities drive the system prediction, the continuous injection of independent inertial measurements serves to stabilize the heading estimate, providing a standard non-linear baseline before introducing landmark corrections.

2.4. Landmark detection and its implementation in the EKF

To counteract the unbounded position drift of the dead-reckoning prediction, an external correction layer using static cylindrical landmarks was integrated into a separate EKF node. The detection pipeline processes raw data from the `/scan` topic, extracting continuous point clusters that match the geometric profile of a cylinder. To achieve this, the algorithm steps through the laser scan array utilizing a structural downsampling factor (*stride*) to optimize runtime performance, while a frame-skipping parameter (*scan_skip*) determines how many consecutive sensor frames are bypassed before executing the next extraction loop. A valid landmark candidate must contain a minimum number of sensor hits (*min_votes*) and fall within a precise geometric window defined by a target radius and an absolute threshold (*tolerance*). Initially, the experimental setup

was designed with three distinct landmarks of varying dimensions – featuring radii of 100 mm, 350 mm, and 600 mm – with the intention of using unique geometric signatures to explicitly resolve the orientation ambiguity inherent to single symmetric structures. However, simulation testing revealed severe edge cases in the extraction logic. The large 600 mm cylinder became statistically indistinguishable from planar wall segments under realistic sensor noise, causing false positives. Conversely, the small 100 mm cylinder was frequently omitted at longer ranges or when spatial point density led to laser scans missing the landmark. Consequently, the multi-size approach was discarded, and the environment was standardized using only the mid-sized 350 mm cylinder. The resulting orientation ambiguity – where an isolated range-bearing measurement matches an infinite set of robot positions on a circle centered at the landmark – is mathematically resolved within the filter framework. Because the EKF continuously maintains a multi-variable state covariance and incorporates heading updates from the IMU, the filter utilizes the robot's motion dynamics over time to uniquely isolate the true coordinate state during the measurement update phase.

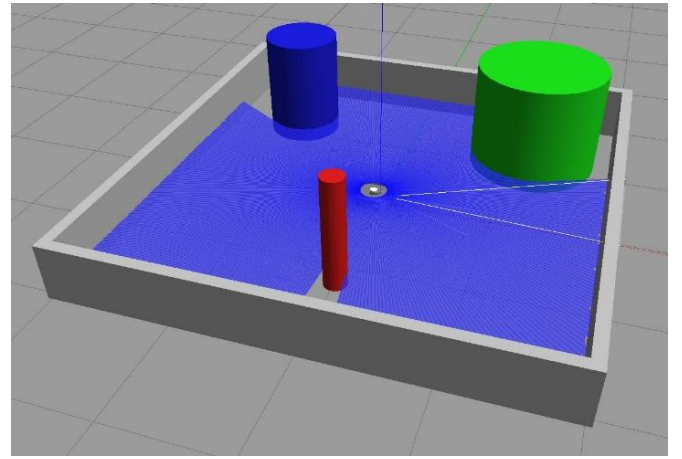


Figure 1: The designed Gazebo world with three distinct cylinders, where the blue one serves as a landmark.

2.5. Particle Filter

The final architecture is a Particle Filter (PF) that models the robot's pose distribution using 1,500 independent hypotheses initialized via a Gaussian distribution. During the prediction phase, each particle is advanced using the non-linear motion equations driven by linear and angular velocity commands from the control input topic. To simulate kinematic drift, Gaussian noise scaled by the elapsed time step is added directly to each particle's coordinates. When a landmark is detected, a measurement update adjusts the particle weights based on a spatial likelihood field. The local range and bearing measurements are projected into global coordinates using each particle's pose hypothesis, and the squared Euclidean distance error to the known landmark location is evaluated. This error is passed through a Gaussian probability density function to compute a matching likelihood that scales the particle's weight. To prevent sample degeneracy, a low-variance resampling step is triggered whenever the effective particle count drops below 50% of the swarm size. Finally, to eliminate discrete trajectory jumps caused by resampling, an exponential smoothing filter (*alpha_smooth*) is applied to the weighted mean. The

orientation smoothing utilizes a wrap-safe vector addition of sine and cosine components, and the final state variances are computed relative to this smoothed estimate before being published.

2.6. Evaluation Methods

To implement a benchmark, all estimation architectures are evaluated under a shared experimental protocol. The robot executes identical test trajectories within the Gazebo world, ensuring that environmental conditions, ground truth paths, and landmark locations remain constant across all trials. To evaluate the filters' handling of angular updates and heading drift, the test trajectory incorporates five points where the robot performs a zero-radius turn on the spot. While the evaluation conditions are unified, minor structural adaptations were required to accommodate the mathematical constraints of specific filters. The KF serves as a simplified baseline, since it is restricted to linear system equations. It operates on a decoupled position-velocity state and is evaluated solely on its position accuracy. Furthermore, while the EKF variants utilize the wheel-encoder feedback from the noisy odometry as their control input, the Particle Filter relies directly on the directional velocity commands */cmd_vel*. This architectural divergence reflects the native properties of the algorithms, testing the PF's capability to sustain localization using forward control inputs versus the EKF's reliance on proprioceptive joint measurements. Performance across all nodes is quantified by 16 experimental runs. The process and measurement noise Q and R are varied using order-of-magnitude scaling factors, utilizing baseline configurations by 0.01 for low noise, 1.0 for standard noise, 100.0 for extreme noise, and an intermediate value of 10.0 to capture non-linear degradation trends. For each run, two primary error metrics are computed against the ground truth frame: the overall Root Mean Square Error (RMSE) for spatial tracking, and the final Euclidean distance error calculated between the absolute terminal point of the ground truth path and the final estimated coordinate of the respective filter. The second metric provides an indication of the absolute offset remaining at the end of the trajectory.

3. RESULTS

To comprehensively test the state estimation nodes, the experimental evaluation is structured into four distinct groups. Each group maintains a constant measurement noise factor R while systematically varying the process noise factor Q across four orders of magnitude (0.01, 1.0, 10.0 and 100.0). The architectures are evaluated using the trajectory's overall Root Mean Square Error (RMSE) for spatial tracking accuracy and the final Euclidean endpoint distance to quantify long-term cumulative drift. All experimental runs were recorded, and the resulting trajectory paths were plotted against the ground truth. As a representative visual reference, Figure 2 illustrates the plotted trajectories for the baseline configuration with minimal noise parameters $Q = 0.01, R = 0.01$. The complete set of trajectory plots for all 16 experimental runs is accessible within the project's Git repository. The tables of this chapter are also visualized and can be found in the appendix.

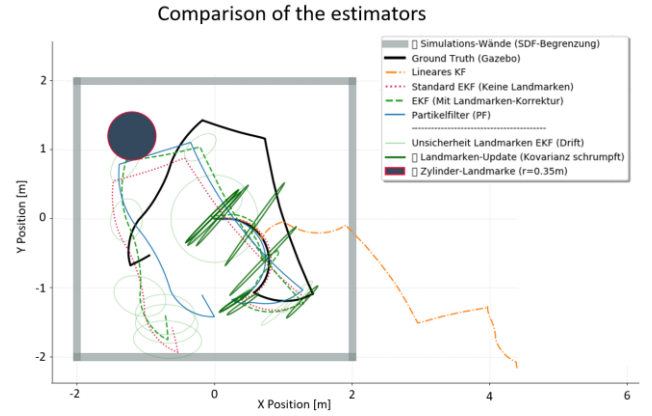


Figure 2: Plotted trajectories of the experimental run with $Q = R = 0.01$.

3.1. Evaluation Group 1: Minimal Measurement Noise with the Factor $R = 0.01$

The first experimental group establishes a baseline with highly confident measurement updates by setting the measurement noise factor R to 0.01.

Table 1: Results of Evaluation Group 1 with minimal R .

Filter	Metric [m]	$Q = 0.01$	$Q = 1.0$	$Q = 10.0$	$Q = 100.0$
KF	RMSE	2.48	3.22	2.27	2.24
	Endpt.	5.85	5.53	4.74	5.22
EKF	RMSE	1.79	1.55	1.82	1.73
	Endpt.	1.11	1.18	0.96	0.83
EKF with LM	RMSE	1.77	1.42	1.56	1.62
	Endpt.	0.88	1.69	1.40	1.34
PF	RMSE	1.83	1.44	1.36	1.32
	Endpt.	1.25	0.58	0.63	0.4

Under these conditions, the linear KF exhibits the highest overall tracking errors, with the final endpoint distance remaining consistently above 4.70 m. Because the linear KF operates on a decoupled linear model and completely omits orientation states, it fails to adapt during the five zero-radius turns on the spot, leading to a permanent coordinate offset. Conversely, the non-linear estimators maintain significantly tighter tracking. For the EKF nodes, increasing the process noise factor forces the filter to rely more heavily on the highly confident IMU yaw updates, reducing the final endpoint distance from 1.11 m to 0.83 m. The Particle Filter achieves the lowest overall tracking error in this group at $Q = 100.0$, yielding an RMSE of 1.32 m and an endpoint distance of just 0.4 m. This demonstrates that forcing a higher particle dispersion allows the swarm to adapt rapidly to sharp orientation changes when the sensor likelihood field is highly localized.

3.2. Evaluation Group 2: Low Measurement Noise with the Factor $R = 1.0$

In the second group, the measurement noise factor is increased to 1.0, introducing a standard level of sensor uncertainty to the update steps.

Table 2: Results of Evaluation Group 2 with low R.

Filter	Metric [m]	Q = 0.01	Q = 1.0	Q = 10.0	Q = 100.0
KF	RMSE	2.50	2.37	2.37	2.41
	Endpt.	5.86	5.56	5.54	5.88
EKF	RMSE	1.79	1.74	1.72	1.67
	Endpt.	1.13	1.11	1.10	1.19
EKF with LM	RMSE	1.70	1.55	1.52	1.52
	Endpt.	0.81	1.20	0.95	1.42
PF	RMSE	1.66	1.57	1.47	1.39
	Endpt.	0.77	1.18	0.52	1.06

As sensor uncertainty rises, the linear KF remains locked in its degraded tracking pattern, showing an average RMSE of approximately 2.40 *m* regardless of *Q* tuning. This confirms that the baseline model's performance is bottlenecked by its lack of kinematic angular projection rather than noise parameterization. The standard EKF exhibits high robustness, keeping its spatial tracking RMSE stable at approximately 1.70 *m*. The integration of landmark detections within the EKF frame provides a noticeable advantage at lower process noise values (*Q* = 0.01), reducing the cumulative tracking drift to an endpoint error of 0.81 *m*. However, as *Q* scales to 100.0, the landmark-enabled EKF undergoes minor degradation, resulting in an increased endpoint error of 1.42 *m*. This indicates that excessive process noise parameters cause the filter to over-correct its position estimates based on single landmark scans, injecting artificial jitter during the on-the-spot rotations.

3.3. Evaluation Group 3: Intermediate Measurement Noise with the Factor R = 10.0

The third experimental group raises the measurement noise factor to 10.0, penalizing the filter corrections and heavily weighting the system's dead-reckoning prediction.

Table 3: Results of Evaluation Group 3 with intermediate R.

Filter	Metric [m]	Q = 0.01	Q = 1.0	Q = 10.0	Q = 100.0
KF	RMSE	2.42	2.48	2.35	2.28
	Endpt.	5.98	5.61	5.79	5.49
EKF	RMSE	1.67	1.70	1.70	1.76
	Endpt.	1.27	1.46	0.90	1.55
EKF with LM	RMSE	1.57	1.60	1.67	1.52
	Endpt.	1.28	1.44	0.90	1.84
PF	RMSE	1.63	1.62	1.45	1.29
	Endpt.	0.72	0.95	0.45	1.39

With highly de-weighted measurement updates, the absolute tracking performance becomes heavily dependent on the synchronization of the motion prediction. For the standard EKF and the landmark-enabled variant, the results converge closely, showing an average RMSE between 1.52 *m* and 1.76 *m*. When *Q* is set to 10.0, both EKF variants present identical metrics, matching at an RMSE of 1.70 *m* and a terminal endpoint distance of 0.90 *m*. At the maximum process noise level (*Q* = 100.0), the landmark-enabled EKF registers its lowest RMSE at

1.52 *m* but simultaneously exhibits its highest final endpoint error at 1.84 *m*. The Particle Filter shows a distinct divergence between its two tracking metrics as the process noise increases. At *Q* = 10.0, the node records a minimized terminal drift, dropping the Euclidean endpoint distance to 0.45 *m*. However, when transitioning to *Q* = 100.0, the performance trends split: the overall spatial tracking reaches its optimal Group 3 value with an RMSE of 1.29 *m*, whereas the final endpoint distance increases significantly to 1.39 *m*. The linear KF shows no major deviation from the previous groups, maintaining a stable tracking error with an average RMSE of 2.38 *m* and terminal offsets consistently above 5.40 *m*.

3.4. Evaluation Group 4: Extreme Measurement Noise with the Factor R = 100.0

The final experimental group simulates severe sensor degradation by setting the measurement noise factor *R* to 100.0, effectively forcing the nodes to rely almost exclusively on their kinematic models.

Table 4: Results of Evaluation Group 4 with extreme R.

Filter	Metric [m]	Q = 0.01	Q = 1.0	Q = 10.0	Q = 100.0
KF	RMSE	2.42	2.44	2.39	2.46
	Endpt.	5.92	5.97	5.86	5.91
EKF	RMSE	1.63	1.73	1.67	1.65
	Endpt.	1.49	1.45	1.45	1.47
EKF with LM	RMSE	1.65	1.74	1.65	1.63
	Endpt.	1.44	1.42	1.35	1.43
PF	RMSE	1.64	1.58	1.57	1.46
	Endpt.	0.95	0.87	0.77	0.77

Under these extreme constraints, the performance of the standard EKF and the landmark-enabled EKF becomes virtually identical. Across all four process noise variations, both non-linear architectures log a stable trajectory tracking RMSE between 1.63 *m* and 1.74 *m*. Their terminal position errors also converge tightly, maintaining a steady Euclidean endpoint distance of approximately 1.45 *m* regardless of whether the landmark node is active. The Particle Filter continues to show scaling improvements in its tracking accuracy as the process noise increases. The overall spatial RMSE decreases approximately linearly from 1.64 *m* at *Q* = 0.01 down to 1.46 *m* at *Q* = 100.0. Similarly, the final endpoint distance improves from 0.95 *m* at the lowest process noise value and stabilizes at exactly 0.77 *m* for both *Q* = 10.0 and *Q* = 100.0. The linear KF presents no structural changes, tracking consistently with a high spatial error (average RMSE of 2.43 *m*) and the largest final endpoint offsets of the group, peaking at 5.97 *m* when *Q* = 1.0.

4. DISCUSSION

The experimental results across the 16 parameter configurations highlight fundamental differences in how parametric, non-parametric, linear, and non-linear state estimation methods handle mobile robot localization. The Kalman Filter demonstrates the strict physical limitations of relying on an oversimplified, decoupled kinematic model.

Because a standard KF cannot resolve trigonometric transformations, it remains completely blind to changes in the robot's heading θ . During the five zero-radius turns on the spot, the linear model continues to project linear velocity commands onto its initial coordinate axes. This structural mismatch produces a permanent, non-recoverable spatial drift, bounding its root-mean-square tracking accuracy to an average RMSE of approximately 2.40 m and a terminal endpoint error exceeding 5.50 m across all evaluations. These results validate why standard linear architectures are insufficient for planar mobile robotics, serving primarily as a degraded baseline. The Extended Kalman Filter (EKF) addresses this limitation by linearizing the motion equations around the current state estimate via Jacobian matrices. The standard EKF displays excellent robustness against sensor noise fluctuations, maintaining a stable tracking profile. However, its accuracy depends heavily on the ratio between process noise Q and measurement noise R . When the measurement updates are highly trusted $R = 0.01$, scaling up the process covariance Q improves terminal tracking by forcing the filter to prioritize active IMU yaw updates over dead reckoning. Conversely, when measurement noise is at least intermediate $(R = 10.0)$, the Kalman gain decreases significantly. This forces the system to rely almost exclusively on proprioceptive wheel encoder data, causing it to integrate unmodeled wheel slippage during the sharp turning points and driving the terminal endpoint drift up to 1.55 m . The integration of custom landmark detection into the EKF framework reveals a delicate performance trade-off. When sensor data is reliable $(R = 1.0)$, exteroceptive landmark updates successfully anchor the robot's Cartesian estimate, yielding a low terminal error of 0.81 m . However, as Q scales up to extreme levels $(Q = 100.0)$, this node undergoes unexpected terminal degradation. Under extreme process covariance settings, the filter over-inflates its state uncertainty between updates. Consequently, the arrival of a single cylinder measurement triggers an aggressive correction vector. This over-correction injects artificial high-frequency jitter during on-the-spot rotations, causing a split behavior where the overall trajectory profile remains bounded (low RMSE) but the final endpoint settles far from the true destination. Under severe sensor degradation $(R = 100.0)$, the Kalman gain for outer sensory information drops to near zero. This entirely mitigates the advantage of the landmark pipeline, causing the landmark-enabled EKF to converge identically with the standard EKF. The Particle Filter (PF) exhibits the highest absolute tracking resilience and handles non-linearities and extreme sensor noise most effectively. Unlike the EKF nodes, which rely on wheel-encoder integration from the `/odom_noisy` topic, the PF propagates its particle swarm using raw control commands from `/cmd_vel`. This structural separation allows the PF to bypass the wheel slippage and surface disturbances simulated by noisy encoders. When measurement noise is severe $(R = 100.0)$, the non-parametric swarm retains a coherent trajectory shape, stabilizing its final endpoint error at exactly 0.77 m compared to the 1.45 m error seen in the EKF variants. Furthermore, the Particle Filter is the only architecture in this evaluation that incorporates a geometric plausibility check during the measurement update. By filtering out radical coordinate jumps and impossible landmark transitions based on the previous state, the PF actively prevents tracking failures. In contrast, the standard and landmark-enabled EKF variants, as well as the linear KF, lack this validation layer. Under high-noise

configurations, this absence allows corrupt measurement updates to induce severe feedback loops, causing the estimated trajectory to diverge realistically to the point where the filter projects the robot driving through structural simulation walls, as showed in figure 3. However, the PF requires careful tuning of its hypothesis dispersion. At intermediate measurement noise $(R = 10.0)$, setting $Q = 10.0$ balances exploration and exploitation, resulting in a minimal terminal error of 0.45 m . Pushing the process noise to an extreme value $(Q = 100.0)$ splits the performance metrics: the average path alignment reaches its optimum with an RMSE of 1.29 m , but the final endpoint drift degrades to 1.39 m . This outcome indicates that while massive noise injection encourages wide exploration and minimizes the average path deviation, it induces destructive particle dispersion during the five angular turning points, disrupting the filter's final convergence accuracy. Additionally, while the integrated exponential smoothing filter ($\alpha_{\text{smooth}} = 0.15$) successfully dampens the discrete trajectory jumps caused by resampling, it introduces a minor temporal lag during abrupt velocity changes.

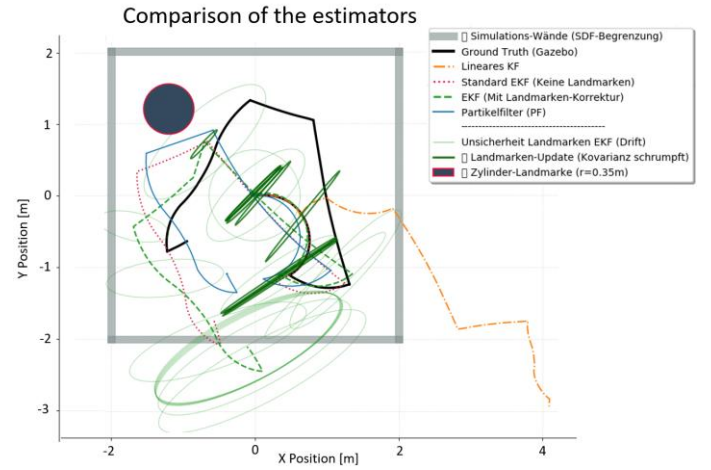


Figure 3: Example run ($Q = 1.0$, $R = 0.01$) where the EKFs estimated the Robot to drive through walls.

5. SUMMARY AND OUTLOOK

This paper presents a comparative benchmarking of four state estimation frameworks – a linear Kalman Filter, an Extended Kalman Filter with and without landmark detection, and a Particle Filter – implemented as ROS2 nodes and evaluated using a TurtleBot3 Waffle within a Gazebo simulation environment. To analyze tracking resilience, the architectures were subjected to an experimental protocol across 16 configurations that systematically scaled the process and measurement noise (Q, R) along a trajectory featuring multiple zero-radius turns. The evaluation highlights that the optimal estimator selection depends heavily on the structural trade-offs of the respective algorithms. The linear KF proved to be insufficient for planar mobile navigation due to its inability to resolve angular state updates. The EKF successfully mitigated these non-linearities, showing high baseline robustness. However, the addition of a custom landmark extraction pipeline introduced a delicate trade-off, bounding drift under low-noise conditions but injecting localized jitter when paired with extreme process noise parameters. The Particle Filter emerged as the most resilient architecture against extreme sensor

degradation. By propagating its swarm via forward control commands rather than proprioceptive joint feedback, the PF successfully bypassed simulated noisy encoders and preserved trajectory coherence even under severe measurement noise. Future work will focus on deploying these state estimation nodes onto a physical TurtleBot to evaluate performance under real-world surface conditions. Additionally, the landmark detection pipeline can be refined and fine-tuned to accommodate environments with multiple simultaneous cylinder landmarks. Finally, integrating a kinematic plausibility check – similar to the one natively utilized by the Particle Filter – into the EKF and KF frameworks represents a vital next step to prevent severe localization failures under extreme noise conditions.

6. REFERENCES

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- [2] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed. Cambridge, MA, USA: MIT Press, 2011.
- [3] T. D. Barfoot, State Estimation for Robotics. Cambridge, UK: Cambridge University Press, 2017.

7. APPENDIX

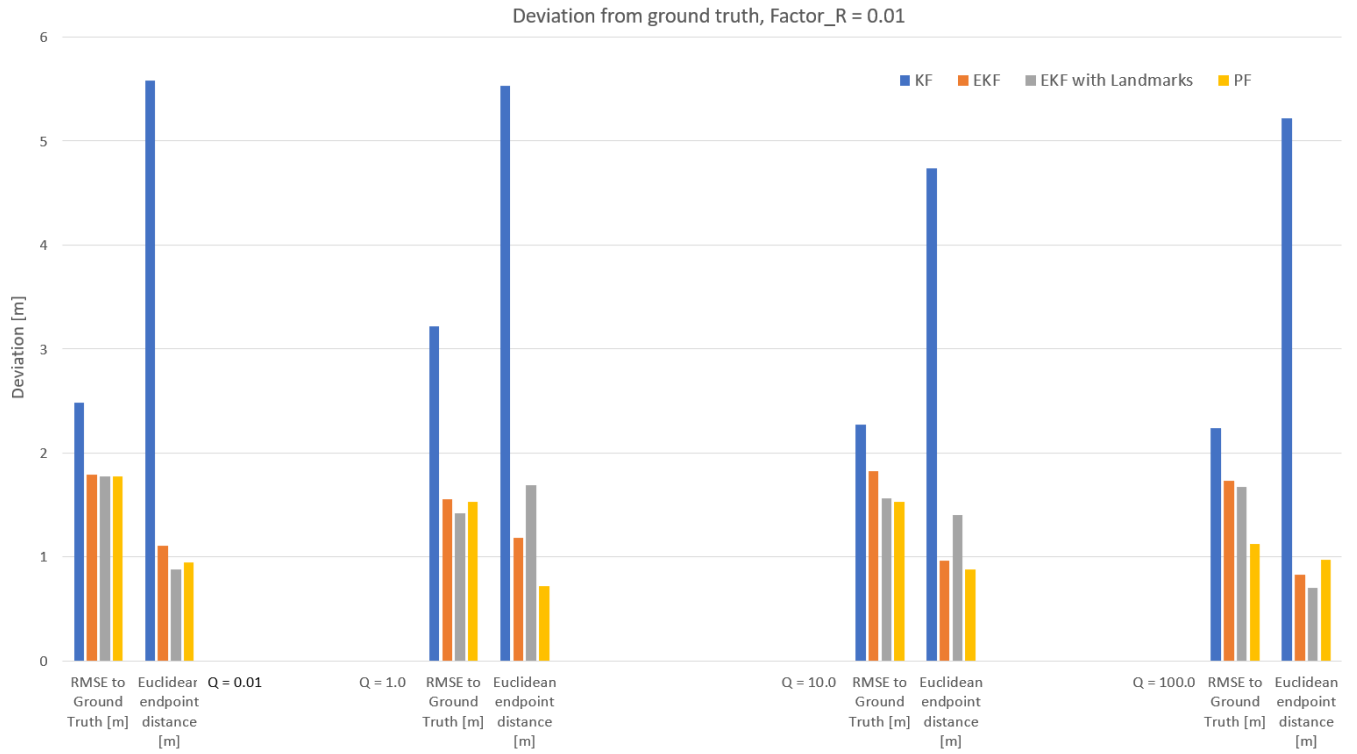


Figure 4: Experimental results of Evaluation Group 1 - R = 0.01.

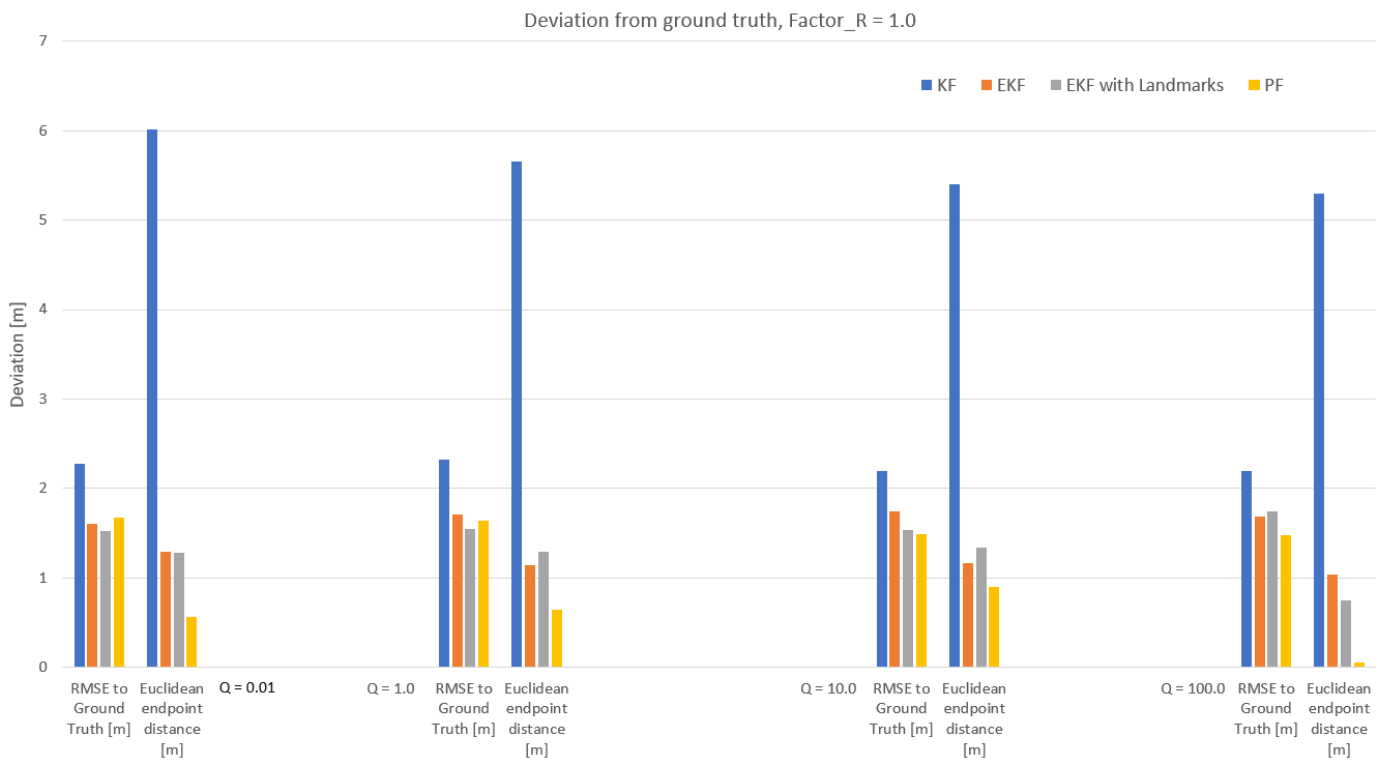


Figure 5: Experimental results of Evaluation Group 2 - R = 1.0.

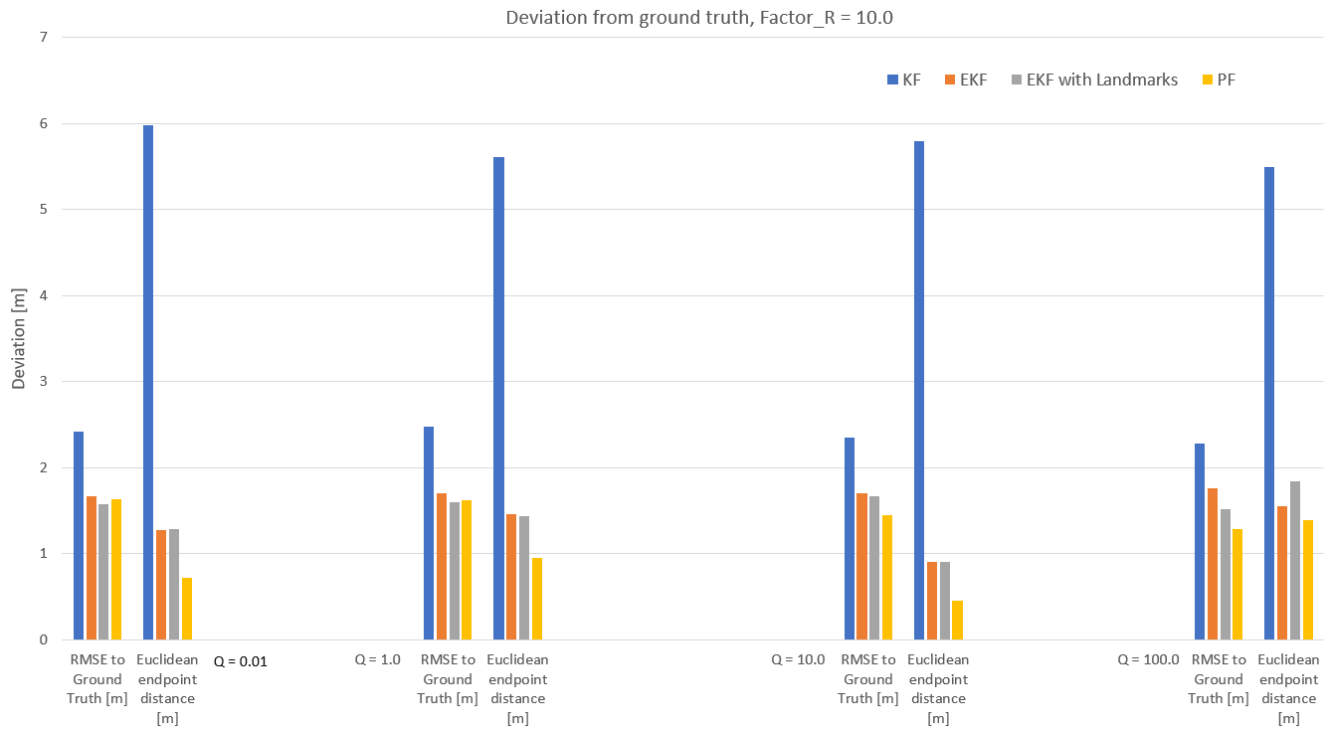


Figure 6: Experimental results of Evaluation Group 3 - R = 10.0.

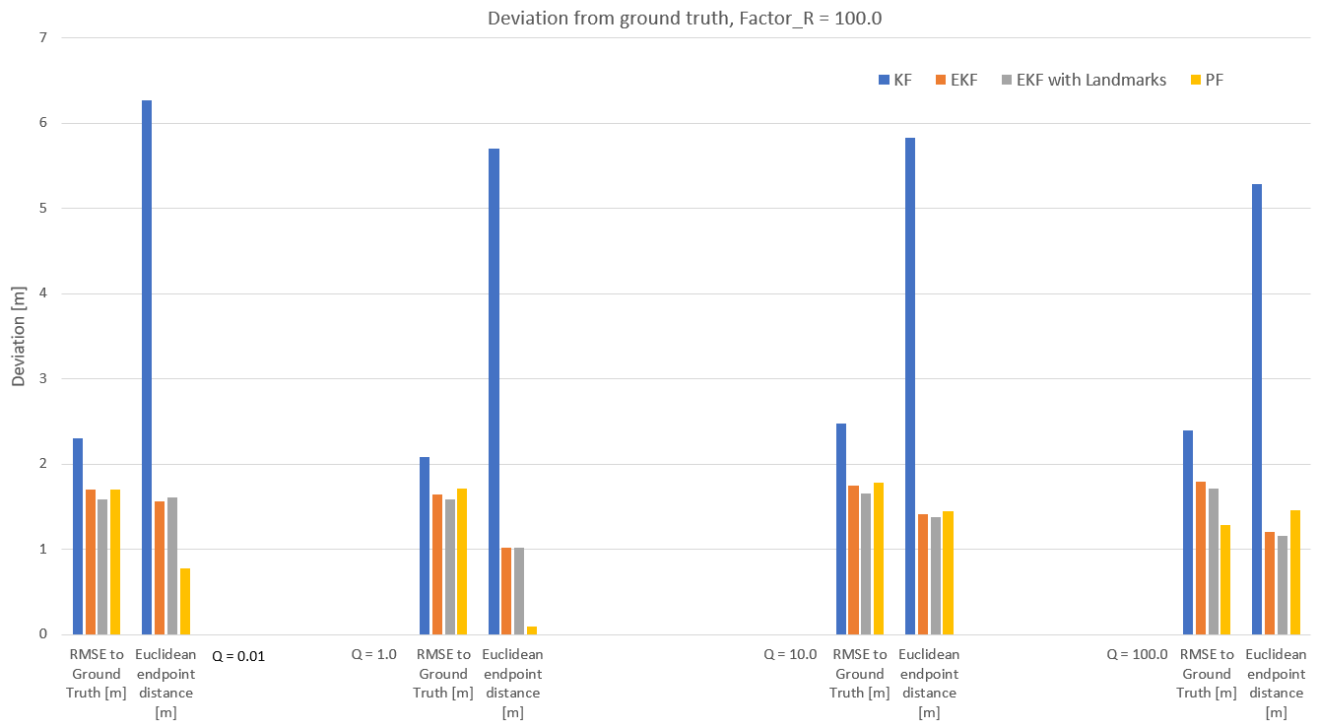


Figure 7: Experimental results of Evaluation Group 4 - R = 100.0.